Label Generation (Explanation)

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Concept Profile (for $r_{\textbf{concept}}$)

* How much of each anomaly contributes to a single window (Ground-Truth)

0.1 Normalizing a feature into $[0, 1]$ (Calibration)

For any nonnegative score $a(t)$, a simple robust normalization is:

$$\tilde{a}(t) = \mathrm{clip}_{[0,1]}\left(\frac{a(t) - q_{0.50}(a)}{q_{0.95}(a) - q_{0.50}(a) + \epsilon}\right)$$

where $q_p(a)$ is the p-quantile over the current window or a reference set.

A) Spike mask $m_{\textbf{spike}}(t)$

Spike evidence should respond to instantaneous amplitude, large derivatives, and peak prominence.

A.1 Hampel amplitude evidence

$$e_{\mathrm{spike},1}(t) = z_x^{\mathrm{ham}}(t)$$

A.2 First-difference (derivative) evidence

Let $\Delta x_t = x_t - x_{t-1}$. Compute a robust z-score on Δx_t:

$$e_{\mathrm{spike},2}(t) = z_{\Delta x}^{\mathrm{ham}}(t) = \frac{|\Delta x_t - \mathrm{med}_t(\Delta x)|}{1.4826 \cdot \mathrm{MAD}_t(\Delta x) + \epsilon}$$

A.3 Peak prominence evidence

Let $p_t \geq 0$ be a peak prominence score. Normalize:

$$e_{\mathrm{spike},3}(t) = \tilde{p}(t)$$

A.4 Combine into raw spike score (Normalize using 0.2)

$$r_{\text{spike}}(t) = \max(\tilde{e}_{\text{spike},1}(t), \tilde{e}_{\text{spike},2}(t)) \cdot \frac{1 + \tilde{e}_{\text{spike},3}(t)}{2}$$

* Either amplitude or deviation is extreme, prominence is a confidence booster

A.5 Soft mask

$$m_{\text{spike}}(t) = \sigma\left(\frac{r_{\text{spike}}(t) - \mu_{\text{spike}}}{s_{\text{spike}}}\right)$$

B) Level shift mask $m_{\textbf{level}}(t)$

Level shift evidence should respond to change-point-like mean shifts and persistence.

B.1 Two-sided mean difference (robust)

Define left and right windows:

$$\mathcal{L}(t) = \{t - w, \ldots, t - 1\}, \quad \mathcal{R}(t) = \{t, \ldots, t + w - 1\}$$

Robust means via median:

$$\mu_L(t) = \text{median}\{x_\tau : \tau \in \mathcal{L}(t)\}, \quad \mu_R(t) = \text{median}\{x_\tau : \tau \in \mathcal{R}(t)\}$$

Scale estimate:

$$s(t) = 1.4826 \cdot \text{MAD}_t(x) + \epsilon$$

Evidence:

$$e_{\text{level},1}(t) = \frac{|\mu_R(t) - \mu_L(t)|}{s(t)}$$

B.2 CUSUM / change statistic

Let $g(t) \geq 0$ be a CUSUM magnitude. Normalize:

$$e_{\text{level},2}(t) = \tilde{g}(t)$$

B.3 Combine and soft mask

$$r_{\text{level}}(t) = \omega_1 \tilde{e}_{\text{level},1}(t) + \omega_2 \tilde{e}_{\text{level},2}(t)$$

$$m_{\text{level}}(t) = \sigma\left(\frac{r_{\text{level}}(t) - \mu_{\text{level}}}{s_{\text{level}}}\right)$$

C) Seasonal break mask $m_{\text{season}}(t)$

Seasonal break evidence should detect a change in seasonal pattern (amplitude, phase, or periodic structure). Uses historical seasonal quantiles—no model fitting required.

C.1 Historical seasonal quantiles

Define a seasonal phase for time t based on domain knowledge (e.g., hour-of-day, day-of-week, or more complex periodic index). For each phase, collect all historical instances:

$$S_{\phi(t)} = \{x_s : s \text{ is in the past and has the same seasonal phase } \phi \text{ as } t\}$$

Compute quantiles from this historical set:

$$Q^{\text{sea}}_{0.05}(t) = \text{quantile}_{0.05}(S_{\phi(t)}), \quad Q^{\text{sea}}_{0.50}(t) = \text{quantile}_{0.50}(S_{\phi(t)}), \quad Q^{\text{sea}}_{0.95}(t) = \text{quantile}_{0.95}(S_{\phi(t)})$$

Evidence 1: deviation from seasonal norm

$$e_{\text{season},1}(t) = \frac{|x_t - Q^{\text{sea}}_{0.50}(t)|}{Q^{\text{sea}}_{0.95}(t) - Q^{\text{sea}}_{0.05}(t) + \epsilon}$$

C.2 Periodogram energy change

Let $P_t(f)$ be local periodogram energy. Compare spectral power around the seasonal frequency f_0 using left/right windows:

$$e_{\text{season},2}(t) = \frac{\left|\text{med}_{\tau\in\mathcal{R}(t)} P_\tau(f_0) - \text{med}_{\tau\in\mathcal{L}(t)} P_\tau(f_0)\right|}{\text{med}_{\tau\in\mathcal{L}(t)} P_\tau(f_0) + \epsilon}$$

A large drop indicates the seasonal oscillation strength has changed.

C.3 Combine and mask

$$r_{\text{season}}(t) = \omega_1 \tilde{e}_{\text{season},1}(t) + \omega_2 \tilde{e}_{\text{season},2}(t)$$
$$m_{\text{season}}(t) = \sigma\left(\frac{r_{\text{season}}(t) - \mu_{\text{season}}}{s_{\text{season}}}\right)$$

D) Contextual deviation mask $m_{\text{context}}(t)$

Contextual deviation means "value is not extreme globally, but is extreme given expected context." Uses quantile regression on recent history to estimate the conditional distribution without fitting an explicit model.

D.1 Conditional range estimation (quantile-based)

From the preceding window $\mathcal{L}(t) = \{t - w, \ldots, t - 1\}$, estimate quantiles:

$$Q_{0.05}^{(t)} = \text{quantile}_{0.05}\{x_\tau : \tau \in \mathcal{L}(t)\}$$

$$Q_{0.50}^{(t)} = \text{quantile}_{0.50}\{x_\tau : \tau \in \mathcal{L}(t)\}$$

$$Q_{0.95}^{(t)} = \text{quantile}_{0.95}\{x_\tau : \tau \in \mathcal{L}(t)\}$$

Deviation from expected range:

$$e_{\text{context},1}(t) = \begin{cases} \dfrac{|x_t - Q_{0.50}^{(t)}|}{Q_{0.95}^{(t)} - Q_{0.05}^{(t)} + \epsilon} & \text{if } x_t \in [Q_{0.05}^{(t)}, Q_{0.95}^{(t)}] \\ \dfrac{\max(|x_t - Q_{0.95}^{(t)}|, |x_t - Q_{0.05}^{(t)}|)}{Q_{0.95}^{(t)} - Q_{0.05}^{(t)} + \epsilon} & \text{otherwise} \end{cases}$$

D.2 Robust z-score on prediction errors

Alternatively, compute error from median baseline:

$$\text{error}_t = x_t - Q_{0.50}^{(t)}$$

$$e_{\text{context},2}(t) = z_{\text{error}}^{\text{ham}}(t) = \frac{|\text{error}_t - \text{med}_t(\text{error})|}{\text{MAD}_t(\text{error}) + \epsilon}$$

D.3 Combine and mask

$$r_{\text{context}}(t) = \omega_1 \tilde{e}_{\text{context},1}(t) + \omega_2 \tilde{e}_{\text{context},2}(t)$$

$$m_{\text{context}}(t) = \sigma\left(\frac{r_{\text{context}}(t) - \mu_{\text{context}}}{s_{\text{context}}}\right)$$

E) Correlation break mask $m_{\textbf{corr}}(t)$ for multivariate

Correlation break measures changes in relationships for multivariate $\mathbf{x}_t \in \mathbb{R}^d$.

E.1 Sliding-window correlation and baseline reference

Compute a sliding-window correlation matrix at time t over a window of width w_c:

$$R_t = \text{corr}_{\text{spearman}}\{\mathbf{X}_{t-w_c+1:t}\}$$

Here we recommend Spearman (rank) correlation for robustness to outliers and nonlinear monotonic relations. Choose a long-term reference correlation matrix R_{ref} (for example the elementwise median of R_t over a reference period, or an exponential moving median).

E.2 Window-to-reference distance (global evidence)

Measure the matrix difference and normalize by matrix size so the score is comparable across dimension d:

$$d_{\text{corr}}(t) = \frac{\|R_t - R_{\text{ref}}\|_F}{\sqrt{d(d-1)}}$$

The denominator $\sqrt{d(d-1)}$ is the RMS normalization of the off-diagonal entries (since a correlation matrix has $d(d-1)$ ordered pairs excluding the diagonal), making d_{corr} approximately scale-invariant in d.

E.3 Node-level localization

To identify which variable(s) drive the change, compute per-node deviations:

$$d_{\text{corr}}^{(i)}(t) = \sum_{j \neq i} |R_t(i,j) - R_{\text{ref}}(i,j)|$$

Large $d_{\text{corr}}^{(i)}$ indicates variable i's relationships changed.

E.4 Calibration and soft mask

Calibrate the distance to a soft mask using robust baseline statistics computed over a reference set of times (e.g., the same period used to form R_{ref}):

$$b_r = \text{median}\{d_{\text{corr}}(t')\}_{t' \in \text{ref}}, \qquad s_r = 1.4826 \cdot \text{MAD}\{d_{\text{corr}}(t')\}_{t' \in \text{ref}} + \epsilon$$

$$m_{\text{corr}}(t) = \sigma\left(\frac{d_{\text{corr}}(t) - b_r}{s_r}\right)$$

For node-level soft masks, apply the same calibration to $d_{\text{corr}}^{(i)}(t)$ (compute node-level baseline medians/MADs if desired).

E.6 Implementation notes

- Use Spearman by default for robustness; Pearson may be used when linear Gaussian assumptions hold.
- Choose w_c to balance frequency resolution and estimation variance (larger w_c gives more stable R_t but slower detection).
- Choose the reference period to reflect the "normal" regime; update R_{ref} occasionally (rolling median) to adapt to slow regime changes.

F) Window-level concept presence (concept profile aggregation)

Because the regression target is a **single window-level utility** $U(W_j)$, we aggregate per-timepoint masks $m_c(t)$ into a **window-level concept profile** $p_{j,c}$. This aligns feature granularity with the label granularity, reduces noise, and yields stable, interpretable coefficients in the detector–concept regression.

Let W_j be a window and $|W_j|$ its length. Define a presence threshold τ_c for concept c.

F.1 Transient concepts (Spike)

For brief, localized anomalies, presence is best captured by a peak statistic (and optionally a normalized count). First compute:

$$p_{j,\text{spike}}^{\max} = \max_{t \in W_j} m_{\text{spike}}(t)$$

$$p_{j,\text{spike}}^{\text{count}} = \#\{t \in W_j : m_{\text{spike}}(t) > \tau_{\text{spike}}\}$$

and the normalized count fraction:

$$p_{j,\text{spike}}^{\text{frac}} = \frac{p_{j,\text{spike}}^{\text{count}}}{|W_j|}.$$

Then define a single scalar spike presence (one feature per concept) as:

$$p_{j,\text{spike}} = \max\left(p_{j,\text{spike}}^{\max},\ p_{j,\text{spike}}^{\text{frac}}\right).$$

(Optionally track mean for logging/debugging:

$$p_{j,\text{spike}}^{\text{mean}} = \frac{1}{|W_j|} \sum_{t \in W_j} m_{\text{spike}}(t).$$

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F.2 Persistent concepts (Level shift, Seasonal break)

For sustained or structural anomalies, we summarize both **average intensity** and **sustained presence**. For $c \in \{\text{level}, \text{season}\}$:

$$p_{j,c}^{\text{mean}} = \frac{1}{|W_j|} \sum_{t \in W_j} m_c(t)$$

$$p_{j,c}^{\text{sustained}} = \frac{\#\{t \in W_j : m_c(t) > \tau_c\}}{|W_j|}.$$

Then define one scalar presence per concept:

$$p_{j,c} = \frac{1}{2}\left(p_{j,c}^{\text{mean}} + p_{j,c}^{\text{sustained}}\right), \qquad c \in \{\text{level}, \text{season}\}.$$

F.3 Mixed concepts (Context deviation, Correlation break)

For concepts that can be either sporadic bursts or sustained shifts, we combine peak and mean. For $c \in \{\text{context}, \text{corr}\}$:

$$p_{j,c}^{\max} = \max_{t \in W_j} m_c(t)$$

$$p_{j,c}^{\text{mean}} = \frac{1}{|W_j|} \sum_{t \in W_j} m_c(t).$$

Then define one scalar presence per concept:

$$p_{j,c} = \frac{1}{2}\left(p_{j,c}^{\max} + p_{j,c}^{\text{mean}}\right), \qquad c \in \{\text{context}, \text{corr}\}.$$

Model–Detector (Capability Profile)

Goal: quantify how well each foundation model m performs on a window W_j as a function of the window's concept profile $p_{j,c}$, by fitting a per-model regression with target $U_m(W_j)$.

1) Define the window-level target utility $U_m(W_j)$

Fix an evaluation protocol. For each model m and each window W_j:

1. Run model m to obtain anomaly scores $\{s_t^{(m)} : t \in W_j\}$.
2. Normalize scores consistently (using statistics fitted on a reference split, not on W_j).
3. Apply a threshold policy to obtain alerts/predictions on W_j.
4. Compute window-level metrics and combine them into a scalar utility.

A typical utility (computed on the evaluation split for W_j) is:

$$U_m(W_j) = \lambda_1 \mathrm{AUPRC}_m(W_j) + \lambda_2 \mathrm{EventF1}_m(W_j) + \lambda_3 \mathrm{NAB}_m(W_j) - \lambda_4 \mathrm{FP/day}_m(W_j) - \lambda_5 \mathrm{Delay}_m(W_j).$$

Threshold calibration (to avoid leakage). Do **not** choose τ to maximize $U_m(W_j)$ on the same window. Instead, for each model m choose a calibrated threshold $\tau_m^\star$ on a separate calibration split (either global across windows or per-series), e.g. by maximizing a calibration utility or by enforcing an alert budget such as FP/day $\leq B$. Then define the target used for learning as:

$$U_m(W_j) \;:=\; U\Big(\tilde{s}^{(m)};\; \tau_m^\star;\; W_j\Big),$$

where $U(\cdot)$ denotes the true metric pipeline applied to window W_j.

2) Construct the concept feature matrix X

For each window W_j, compute the concept profile $p_{j,c} \in [0,1]$ for each concept $c \in \mathcal{C}$ (e.g., spike/level/season/context/corr). Stack these into a design matrix:

$$X \in \mathbb{R}^{N \times C}, \qquad X_{j,c} = p_{j,c},$$

where N is the number of windows and $C = |\mathcal{C}|$.

3) Fit a regression per model m (capability profile)

For each model m, fit a regularized linear regression that predicts the window-level utility $U_m(W_j)$ from the concept profile:

$$U_m(W_j) = \beta_{m,0} + \sum_{c \in \mathcal{C}} \beta_{m,c}\, p_{j,c} + \epsilon_j.$$

In vector form, letting $y_j^{(m)} = U_m(W_j)$:

$$y^{(m)} = \beta_{m,0}\mathbf{1} + X\beta_m + \epsilon, \qquad \beta_m \in \mathbb{R}^C.$$

Faithfulness reward via structured extraction

Given a window W_j with concept profile $p_{j,c}$ and a chosen model m, let the LLM produce an explanation text E_j.

1) Extract structured claims from E_j

We map free text E_j into a fixed schema over the concept set $\mathcal{C}$:

- **Concept mentions** ("what happened"): $\hat{q}_{j,c} \geq 0,\ \sum_c \hat{q}_{j,c} = 1$.
- **Justification attributions** ("why model m"): $\hat{a}_{j,c} \geq 0,\ \sum_c \hat{a}_{j,c} = 1$.

In practice, $\hat{q}_{j,c}$ and $\hat{a}_{j,c}$ are obtained by a constrained extractor (LLM or rules) that (i) only allows concepts in $\mathcal{C}$, (ii) outputs normalized weights, and (iii) optionally returns explicit $(\text{model}, \text{concept})$ justification links.

2) Concept faithfulness term r_{concept}

Normalize the concept profile:

$$\tilde{p}_{j,c} = \frac{p_{j,c}}{\sum_{c'} p_{j,c'} + \epsilon}.$$

Compute concept faithfulness by cosine similarity:

$$r_{\text{concept}} = \frac{\sum_c \tilde{p}_{j,c}\hat{q}_{j,c}}{\sqrt{\sum_c \tilde{p}_{j,c}^2}\sqrt{\sum_c \hat{q}_{j,c}^2} + \epsilon} \in [0, 1].$$

3) Capability (model-justification) faithfulness term r_{model}

Using the capability regression coefficients $\beta_{m,c}$, define per-concept positive contributions for window W_j:

$$g_{j,m,c} = \beta_{m,c}\, p_{j,c}, \qquad g_{j,m,c}^{+} = \max(g_{j,m,c}, 0),$$

and normalize them into a "grounded justification" distribution:

$$\pi_{j,m,c} = \frac{g_{j,m,c}^{+}}{\sum_{c'} g_{j,m,c'}^{+} + \epsilon}.$$

Compare the explanation attributions $\hat{a}_{j,c}$ to $\pi_{j,m,c}$ via cosine similarity:

$$r_{\text{model}} = \frac{\sum_c \pi_{j,m,c}\hat{a}_{j,c}}{\sqrt{\sum_c \pi_{j,m,c}^2}\sqrt{\sum_c \hat{a}_{j,c}^2} + \epsilon} \in [0, 1].$$

4) Combined faithfulness reward

Define the combined faithfulness reward:

$$r_{\text{faithfulness}} = \text{clip}_{[0,1]}\Big(r_{\text{decision}} + \alpha\, r_{\text{confidence}} + \beta\, r_{\text{model}} + \gamma\, r_{\text{concept}}\Big).$$

Here r_{decision} rewards selecting the correct action/model (teacher or oracle), and $r_{\text{confidence}}$ rewards calibrated confidence; r_{model} and r_{concept} evaluate whether the explanation is grounded in the capability and concept profiles, respectively.